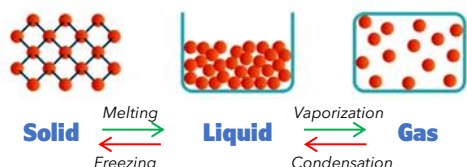


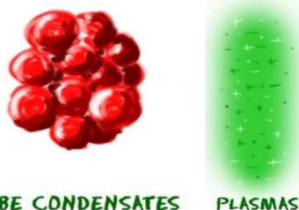
States of Matter *Continuation...*

Matter occurs in three different forms depending on the conditions of the surroundings. For example, water, which is one of the most common substances we will ever see, can occur in different forms depending on the temperature. At normal temperature, it is liquid in form. Raising the temperature above its boiling point, it will be in gaseous form. Lowering its temperature below its freezing point, it will be solid ice. Thus, matter can be transformed from one form to another and this will involve the transfer of energy in the form of heat.

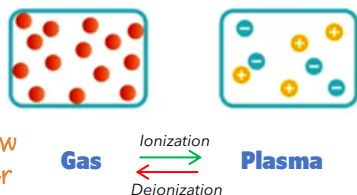
Excel Review Center



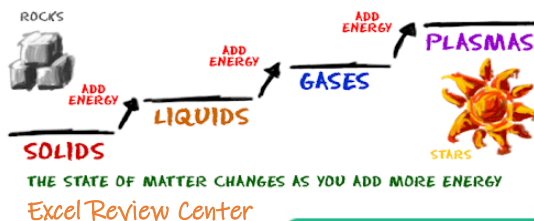
Two Special States



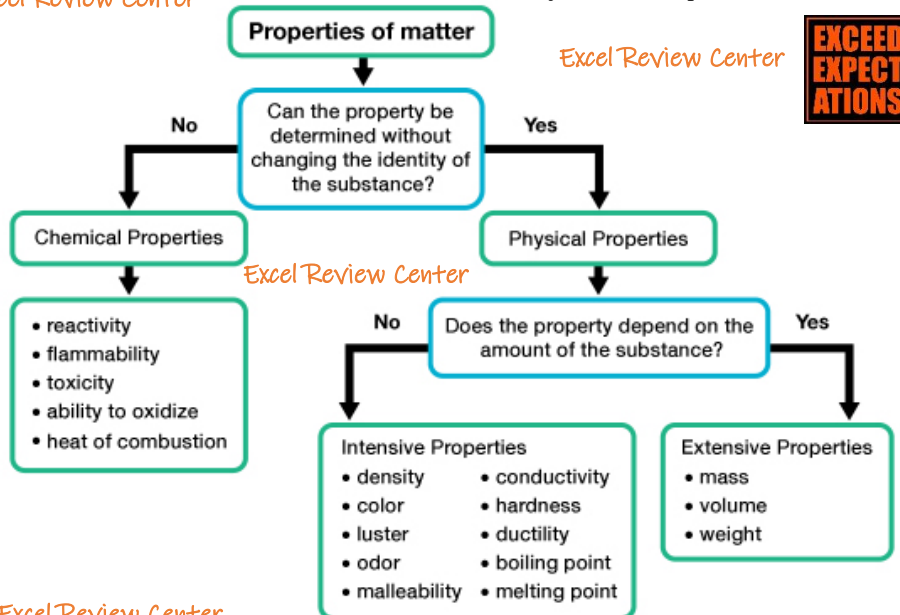
- Plasma** – similar to gases except that it is made up of free electrons and ions from the element. Under certain conditions, gases will behave like plasma. For instance, neon signs are glass tubes filled with gas. When electricity flows through the tube (e.g. the light is turned on), the electricity charges the gas and creates plasma inside the tube.
- Bose-Einstein Condensate (BEC)** – the newest state of matter as it was discovered only in 1995 by Cornell and Weiman, although Satyendra Bose and Albert Einstein predicted this state of matter in the 1920's. It is the opposite of plasma so this state of matter must be supercooled and will occur a few billionths temperature within the absolute zero temperature.



Excel
Review
Center



Excel Review Center



Excel Review Center

Excel Review Center

Physical Properties	Chemical Properties
Mass	Conductivity
Color	Malleability
Odor	Reactivity
Size	Flammability
Freezing Point	Toxicity
Boiling Point	Radioactivity
Density	Oxidation

Extensive Properties	Intensive Properties
Mass	Boiling Point
Volume	Freezing Point
Length	Density
Weight	Temperature
Area	Pressure
Width	Melting Point
Energy	Color

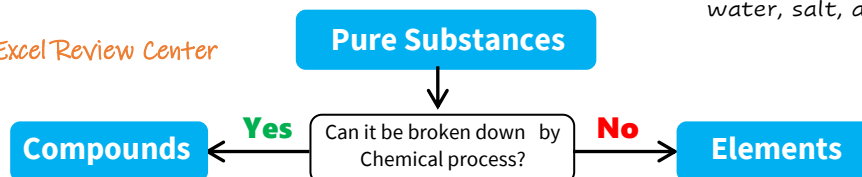
Measurable Properties

- Extensive properties** – these are properties that depend on the amount or on the “extent” of substance or matter being studied; extensive properties are additive which means that more matter equates to a higher value of the extensive property.
- Intensive properties** – these are properties that are independent on the amount of substance or matter being studied; intensive properties are not additive which means that regardless of the amount of substance, these properties have a fixed value.

Excel Review Center

Pure substances are matter that have definite composition and properties. It cannot be separated into two or more substances by physical means. Examples: silver, gold, oxygen, water, salt, ammonia

Excel Review Center



A **compound** is combination of two or more elements chemically combined in fixed proportions. It can be separated by chemical means into its respective components.

Excel
Review
Center

An **element** is a pure substance that cannot be broken down into a much simpler substance by chemical means. Normally, elements are symbolized by one or two letters in which the first letter is always capitalized.